Day-3

1.Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.

DNS Server Setup on Windows

1. Downloaded and installed Simple DNS Plus on Windows.
2. Opened Simple DNS Plus and created a new primary DNS zone named musicmania.in.
3. Added an A record for musicmania.in pointing to 192.168.1.10.
4. Added another A record for www.musicmania.in pointing to 192.168.1.10.
5. Saved the configuration and started the DNS service.
6. Opened Command Prompt and executed nslookup www.musicmania.in 127.0.0.1.
7. Verified that the domain name resolved to the configured IP address.

Result: A DNS zone named musicmania.in was configured with the required DNS records to resolve the domain to a local IP address.

2. Add an A record in your DNS server configuration to map www.musicmania.in to the IP address 192.168.1.50, then use the nslookup or dig command to verify the mapping works.

Steps to Add and Verify an A Record

1. Opened Simple DNS Plus on Windows and selected the musicmania.in DNS zone.
2. Added a new A record with the hostname www and IP address 192.168.1.50.
3. Saved the DNS configuration and ensured the DNS service was running.
4. Opened Command Prompt and executed the following command:

nslookup www.musicmania.in 127.0.0.1

5. Verified that the DNS lookup returned the IP address 192.168.1.50.

Result: The A record was configured to map www.musicmania.in to 192.168.1.50, and the mapping was verified using nslookup.

3. Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.

Steps to Create and Verify a CNAME Record

1. Opened Simple DNS Plus and selected the musicmania.in DNS zone.
2. Added a CNAME record for play.musicmania.in pointing to www.musicmania.in.
3. Saved the DNS configuration.

4. Opened Command Prompt and ran nslookup play.musicmania.in 127.0.0.1.
5. Verified that play.musicmania.in resolved to the IP address 192.168.1.50.
6. Result: The CNAME record successfully mapped play.musicmania.in to www.musicmania.in, which resolved to 192.168.1.50

   4.Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig.r

   DNS Troubleshooting for musicmania.in

   Opened Simple DNS Plus and checked the DNS zone configuration for musicmania.in. Verified that the A record points to the correct IP address, 192.168.1.50.

   Checked whether the DNS service was running and restarted it if necessary.

   Opened Command Prompt and tested the DNS resolution using:

   nslookup musicmania.in 127.0.0.1

   Checked the DNS cache and cleared it using:

   ipconfig /flushdns

   Tested network connectivity to the DNS server using:

   ping 127.0.0.1

   Verified that the DNS server was listening on port 53 and checked the configured DNS server address.

   Corrected any incorrect DNS records or configuration settings, saved the changes, and ran nslookup again to verify the result.

   Result: The DNS configuration and service were checked, incorrect records were corrected if necessary, and DNS resolution was tested again using nslookup